

Research Plan - QIntern 2026

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1. Project Title

Dynamic Entanglement Weighted Pruning for QFL-Based Supply Chain Risk Unlearning

2. Project Overview

This project builds a practical framework that combines **Quantum Federated Learning (QFL)** with **machine unlearning** – the ability to remove a specific client’s data influence from a shared model without retraining from scratch. We apply this to **supply-chain risk forecasting**, where clients (e.g., logistics firms) may legally request data deletion after model training under regulations such as GDPR.

The core idea is straightforward: instead of retraining the entire quantum model when a client wants their data forgotten, so we identify the circuit parameters most responsible for memorising that client’s data and *prune* them. To decide which parameters to prune, we combine two signals:

- **Quantum Fisher information** [1] – measures how sensitive parameter is to data changes; a high diagonal Fisher entry means the parameter “remembers” more.
- **Entanglement weight** – measures how strongly a quantum gate entangles qubits; gates with low entanglement are safer to remove without harming overall model utility.

Combining these two signals gives a *pruning score* s_k for every circuit parameter. Parameters below a threshold τ are pruned, effectively erasing the forgotten client’s influence while preserving accuracy for all remaining clients. The entire implementation uses **Qiskit** [2] and runs on classical simulators; no quantum hardware is required.

Think of it like this: the quantum model is a network of adjustable knobs (parameters), and forgetting a client means carefully turning off only the knobs that “learned” from their data, leaving everything else untouched. This project is, therefore, a great entry point for interns interested in the intersection of quantum computing, privacy law, and real-world logistics, no Prior experience in all three areas is needed, just curiosity and a willingness to learn.

3. Expected Background and Preparation

Interns are expected to have a solid understanding of basic quantum computing principles (qubits, gates, circuits), linear algebra, and familiarity with simple quantum algorithms. A working knowledge of Python and some experience with any quantum programming framework are recommended. As preparation, interns are encouraged to skim the relevant background papers [3–5] and practice writing simple quantum programs using Qiskit tutorials [2].

4. Six-Week Execution Plan

Week	Milestones
Week 1	Study QFL and machine unlearning fundamentals. Set up the Qiskit environment [2] and build a simple variational quantum circuit (VQC) for binary classification on toy data.
Week 2–3	Implement a 3-client QFL simulator using FedAvg-style gradient aggregation [3]. Train a VQC-based risk predictor on synthetic supply-chain data (delay-risk, demand-risk, logistics-risk).
Week 4	Implement the pruning-score computation: parameter-shift rule for Quantum Fisher matrix diagonal estimation [1], and entanglement-entropy estimation (concurrence or von Neumann entropy) for each two-qubit gate.
Week 5	Run a full forget experiment on one client. Compare four baselines: full retrain, fine-tune only, random pruning, and entanglement-weighted pruning [4, 5]. Record model utility and forgetting metrics.
Week 6	Write up results, clean and document the notebook, and prepare an open-source release. Optionally draft an arXiv submission.

5. Method Summary

Each client i runs a parameterised quantum circuit $U(\vec{\theta}_i)$ and minimises a risk-expectation loss:

$$\mathcal{L}_i = \langle \psi(\vec{\theta}_i) | \hat{H}_i^{\text{risk}} | \psi(\vec{\theta}_i) \rangle.$$

When client j issues a forget request, we compute a **pruning score** for every parameter θ_k :

$$s_k = w_{\text{ent}}(k) \cdot \mathcal{F}_{kk}^{(j)},$$

where $w_{\text{ent}}(k)$ is the entanglement weight of the gate acting on θ_k and $\mathcal{F}_{kk}^{(j)}$ is its diagonal Quantum Fisher entry estimated via the parameter-shift rule [1] on client j 's data. Parameters with $s_k < \tau$ are pruned; the remaining parameters may be briefly fine-tuned on the client's retained data.

6. Deliverables

- A well-documented **Jupyter notebook** implementing the full QFL unlearning pipeline in Qiskit [2], submitted as an open-source contribution.
- (*Optional*) An **arXiv paper** comparing entanglement-weighted pruning against full-retrain, fine-tune, and random-pruning baselines [4, 5], co-authored with the mentor and interested interns.

7. Guidance and Extension

I will be actively involved in both the theoretical and implementation aspects of the project, providing regular support via Slack or Discord to address any queries. For Qiskit-specific issues, the core Qiskit community and documentation [2] are available for additional assistance. A short introductory presentation can be arranged if needed. Weekly progress meetings will be held to review milestones, with additional meetings upon request. If interns are interested in extending the work beyond the project duration, we can collaboratively pursue an arXiv publication comparing the implemented entanglement-weighted pruning approach with alternative unlearning techniques [4, 5], including performance benchmarks, resource estimates, and application case studies. Intern-driven ideas for further development are highly encouraged.

References

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- [2] Qiskit contributors. Qiskit: An open-source framework for quantum computing, 2024. URL <https://doi.org/10.5281/zenodo.2573505>.
- [3] H. B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. Agüera y Arcas. Communication-efficient learning of deep networks from decentralized data. In *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics (AISTATS)*, volume 54 of *Proceedings of Machine Learning Research*, pages 1273–1282, 2017. URL <https://proceedings.mlr.press/v54/mcmahan17a.html>.
- [4] Junjian Su, Runze He, Guanghui Li, Sujuan Qin, Zhimin He, Haozhen Situ, and Fei Gao. From membership-privacy leakage to quantum machine unlearning, 2025. URL <https://arxiv.org/abs/2509.06086>.
- [5] Carla Crivoi and Radu Tudor Ionescu. Machine unlearning in the era of quantum machine learning: An empirical study, 2024. URL <https://arxiv.org/abs/2512.19253>.